

AI Infrastructure Resource Ledger: The Reality Check

Thursday, 5. March 2026

Engineer(s): Jay Vakil

Date: 5 Mar 2026

Project Title: PLA: proximity-tokenized VLAs for manipulation in clutter,

Phase 1: The Calendar (Working Backward)

If you are still writing your training loop on your Drop-Dead Date, your project will fail.

1. Final Demo / Presentation Date: Finals week
2. Days required for Evaluation & Report: 14 days.
3. Estimated Training Run Duration: 14-28 days.
4. DROP-DEAD TRAINING START DATE: _____ (Subtract Line 2 & 3 from Line 1)

Phase 2: The Stack & Compute Target

1. Base architecture backbone, include encoders etc., add as many lines as you need:
Octo (for different modalities), transformer, CVAE
Param Count: 0.6 Billion
2. Target Compute Hardware:
☒ Local / Lab Node
☐ Cloud Rental (e.g., Lambda/AWS)
3. Specific GPU(s): 1 4090
Total VRAM Available: 24 GB

Phase 3: The Memory Math (The VRAM Ledger)

Reference: 1B Parameters $\approx$ 2GB (Weights) + 2GB (Gradients) + 12GB (AdamW Optimizer States)

1. Model Weights (BF16): _____ GB
2. Gradients (BF16): _____ GB
3. Optimizer States: _____ GB
4. KV Cache & Activations Reserve: _____ GB
5. TOTAL REQUIRED VRAM: _____ GB

The Strategy: Is your Total Required VRAM > Total Available VRAM?

☐ Yes ☒ No

If Yes, what is your exact mitigation strategy? ☐ FSDP / ZeRO-3 (Sharding across multiple GPUs)

☐ LoRA / QLoRA (Freezing base weights, avoiding optimizer state explosion)

☐ I am shrinking my model selection.

☐ Something else (explain):

Phase 4: The Data Ingress (The I/O Bottleneck)

1. Dataset Size: 100 GB / 1000Total Samples

2. Format:

☒ Raw JPEGs/PNGs/trajectories/etc. in a folder (High I/O Risk)

☐ WebDataset / Sharded .tar

☐ Something else

3. Storage Location:

☒ Local NVMe SSD

☐ Network Drive

☐ S3/GCS Bucket

☐ Something else

If using raw images or other large data quantities over a network or standard SSD, how will you prevent your dataloader from starving your GPUs?

Phase 5: Red Team Audit (Partner Review)

Reviewing Engineer: _____

Look at your partner's ledger. Find the fatal flaw. Are they underestimating training time? Do they have a 40GB model fitting into 24GB of VRAM? Are they trying to load 2 million JPEGs sequentially?

The Fatal Flaw / Biggest Risk:

Phase 6: The 72-Hour Contract

What exact, granular engineering task will you complete in the next 72 hours to unblock these risks? (e.g., "Write dummy dataloader script to verify throughput," "Run a 10-step overfitting batch to check VRAM usage.")

Deliverable by Monday: Collect a set of data and train an action prediction model. The model is just a basic behavior cloning policy which is a safe deliverable.